

Kinematic–RCS Feature Fusion for Maritime Vessel Classification: Temporal Behavioural Signatures

Break the RCS-Only Accuracy Ceiling

Research Question 6: Does fusing temporally-aggregated radar kinematic signatures with range-invariant RCS features surpass the classification ceiling of either feature family alone?

Vijay Kumar V

Abstract—Papers 1–5 of this series established that range-invariant radar cross section (RCS) features achieve 84.0–84.8% four-class vessel classification accuracy for Gradient Boosting Trees and 77.7–78.3% for Hybrid Quantum-Classical Neural Networks (HQNN). The residual error in the RCS-only regime is dominated by the Cargo/Tanker pair, which share overlapping hull geometry despite representing distinct vessel archetypes. This paper introduces a *kinematic augmentation* that adds four temporally-aggregated behavioural descriptors — instantaneous speed-over-ground, 15-minute mean speed, 15-minute speed standard deviation, and 15-minute course-over-ground standard deviation — to the five RCS features from Paper 5, forming a nine-dimensional kinematic-RCS fused descriptor. On a 300 samples-per-class balanced corpus of 91,205 cleaned coastal X-band radar detections (Fishing, Tug, Cargo, Tanker), Gradient Boosting Trees achieve 94.4% accuracy and 94.4% macro F1 — the highest result in this series and a 10.4 percentage-point improvement over the RCS-only baseline. XGBoost achieves 93.2% (+8.4 pp). HQNN-8q achieves 84.2% (+5.9 pp), now equalling the prior classical ceiling, while HQNN-5q achieves 81.7% (+4.0 pp). The fusion works because RCS features encode *what the vessel looks like* (geometry, reflectivity) while kinematic features encode *how the vessel moves* (transit speed and course stability) — two physically orthogonal information channels that are conditionally nearly independent given vessel class. Cargo and Tanker vessels, while geometrically similar at radar range resolution, differ in their kinematic regimes: tankers maintain exceptionally steady courses on deep-sea lanes, while cargo vessels exhibit greater course variance during port approaches and coastal transits. Course-over-ground standard deviation (σ_ψ) is identified as the dominant kinematic discriminator via EDA (Cohen’s $d = 0.357$) and confirmed in the fused model by feature ablation. Classical ensemble methods scale substantially better than HQNN architectures to the expanded nine-dimensional feature space, suggesting that the quantum advantage in this problem domain lies in calibration and uncertainty quantification rather than accuracy maximisation when training-set size is constrained.

Index Terms—maritime vessel classification; radar; kinematic features; RCS features; feature fusion; temporal aggregation; gradient boosting; XGBoost; quantum machine learning; HQNN; PennyLane; behavioural signatures; Cargo-Tanker separation; course-over-ground variance; coastal surveillance

V. Kumar is an Independent Researcher and Innovator, Bangalore, India. E-mail: vinayitsv14@gmail.com.

This work was self-funded by the author as an independent research contribution with no institutional or corporate affiliation.

I. INTRODUCTION

MARITIME vessel classification from terrestrial radar faces a fundamental challenge: the radar return from a vessel encodes both the electromagnetic interaction with the hull (size, shape, material) and the kinematic state of the vessel (speed, course, manoeuvrability). These two channels of information are measured through different physical pathways and, crucially, are largely independent given the vessel’s type. A radar echo at a single instant captures hull geometry; the sequence of echoes over a vessel’s transit captures its behavioural pattern. Using only one channel leaves discriminative information on the table.

Papers 1–4 of this series established a rigorous baseline using five raw geometric features (azimuth, peak amplitude, total amplitude, down-range extent, cross-range extent in metres) and demonstrated that Gradient Boosting Trees achieve 90.0% accuracy on five vessel types when trained on 300 balanced samples per class [1]. Paper 5 introduced range-invariant RCS engineering — transforming raw amplitudes via the monostatic radar range equation — and achieved 84.0–84.8% on four classes with five physically grounded features, while demonstrating that Hybrid Quantum-Classical Neural Networks (HQNN) achieve substantially better probability calibration than classical ensemble methods [5].

Despite these advances, the *Cargo-Tanker boundary* has remained the dominant classification bottleneck. Both vessel types operate in constrained deep-sea lanes, both have large hulls with broad radar footprints, and their length-to-beam ratios overlap substantially. Paper 4 identified this as the hardest pair in the four-class problem [4], and Paper 5 confirmed that the RCS-only confusion rate remains around 25–30% in either direction.

This paper tests whether the kinematic channel — available in any coherent radar that tracks detections over time — provides the complementary information needed to break this ceiling. Specifically, we augment the five RCS features with four 15-minute rolling statistics of speed and course behaviour, forming a nine-dimensional *kinematic-RCS* descriptor, and evaluate the same four classifier architectures as Papers 2–5 (GBT, XGBoost, HQNN-5q, HQNN-8q) on the same balanced

experimental protocol.

Principal contributions of this paper:

- 1) A kinematic-RCS fused descriptor combining two physically orthogonal information channels from standard coastal radar outputs, requiring no additional hardware or sensing modality.
- 2) Experimental demonstration that GBT achieves 94.4% four-class accuracy on the fused descriptor — a 10.4 pp improvement over the RCS-only result of Paper 5 and the highest accuracy reported in this series.
- 3) Evidence that course-over-ground standard deviation (σ_ψ) is the dominant kinematic discriminator, with a Cohen's $d = 0.357$ Cargo-versus-Tanker effect size confirmed by both EDA and feature ablation.
- 4) Characterisation of asymmetric scaling: classical ensemble methods gain 8–10 pp from kinematic augmentation while HQNN architectures gain 4–6 pp, establishing that the quantum advantage in this domain resides in calibration rather than in accuracy when training data is limited to 300 samples per class.
- 5) Analysis of the residual error structure after fusion: the dominant confusion shifts from Cargo-Tanker (geometric overlap) to Fishing-Tug (behavioural overlap), indicating that the fusion resolves the RCS-only bottleneck but reveals the next limiting physical ambiguity.

The remainder of this paper is organised as follows. Section II summarises the relevant background. Section III derives the kinematic features and justifies their physical discriminability. Section IV describes the dataset and experimental setup. Section V presents quantitative results. Section VI discusses the mechanisms and implications. Section VII concludes.

II. BACKGROUND

A. RCS Features from Paper 5

Paper 5 derived five range-invariant features from the monostatic radar range equation [6]. Received power P_r follows:

$$P_r = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 R^4} \quad (1)$$

where P_t is transmit power, G antenna gain, λ wavelength, σ true RCS, and R slant range. Taking logarithms and isolating σ :

$$\log \hat{\sigma}_p = \log A_p + 4 \log R + C_{\text{sys}} \quad (2)$$

where A_p is the per-detection peak amplitude in system units and C_{sys} is a system constant absorbed into the feature (since classifiers are trained on the same radar). Similarly for total amplitude:

$$\log \hat{\sigma}_t = \log A_t + 4 \log R + C_{\text{sys}} \quad (3)$$

Three additional features encoding shape, concentration, and size:

$$\rho = A_p/A_t \quad (\text{RCS concentration ratio}) \quad (4)$$

$$\xi = d_R/d_A \quad (\text{aspect ratio, down/cross-range}) \quad (5)$$

$$\log A = \log(\pi d_R d_A / 4) \quad (\text{log footprint area}) \quad (6)$$

where d_R is the down-range gate extent and d_A the azimuth extent in metres. Paper 5 showed that GBT on these five features achieves 84.0%, with feature importance dominated by aspect ratio ($\sim 25\%$) and log footprint area ($\sim 23\%$).

B. Temporal Track Features in Radar Systems

Modern pulse-Doppler maritime surveillance radars routinely maintain vessel tracks by associating detections across successive scans. The track processor typically estimates and reports the vessel's *speed over ground* (SoG, derived from successive Doppler measurements or position differencing) and *course over ground* (CoG, derived from successive azimuth and range estimates). These quantities are reported per detection and can be aggregated over a rolling temporal window to produce behavioural statistics that are stable over seconds to minutes of transit.

The key advantage of temporal aggregation is noise reduction: a single scan's SoG estimate may be noisy due to target fading or sea clutter, but the 15-minute mean converges to the true transit speed. More importantly, the *variance* of SoG and CoG over a window characterises the vessel's *behavioural regime* — whether it is maintaining a steady transit (low variance) or manoeuvring (high variance) — in a way that no single-scan measurement can.

C. Vessel Behavioural Archetypes

Maritime vessel types follow physically motivated kinematic archetypes:

- **Cargo (Type 70):** Operates primarily on commercial sea lanes between ports. Maintains a constant course during the transit phase (CoG variance low). Speed varies by vessel size and load (10–18 kts typical). Port approach involves gradual deceleration.
- **Tanker (Type 80):** Very similar to Cargo in transit regime. However, Very Large Crude Carriers (VLCCs) and product tankers on fixed routes tend to maintain particularly steady courses enforced by Traffic Separation Schemes. Speed slightly lower than comparable Cargo (12–15 kts typical for large tankers).
- **Fishing (Type 30):** High behavioural variability. Fishermen alter course frequently to locate fish schools, slow to trolling speeds (1–3 kts), and accelerate between grounds. CoG variance and SoG variance are highest of all four classes.
- **Tug (Type 52):** Operates at low speeds (4–8 kts when towing). CoG variance is moderate to high due to towing

manoeuvres. May appear stationary for extended periods when assisting at berth.

This classification of kinematic regimes motivates the hypothesis that temporal speed and course statistics can supplement RCS geometry to resolve ambiguities that persist in the RCS-only feature space.

D. HQNN Architecture

Hybrid Quantum-Classical Neural Networks [8], [9] used in this series embed classical features into quantum states via angle encoding, process them through parameterised quantum circuits (PQCs) with entangling layers, and pass the quantum measurement outcomes to a classical output network. For n qubits and L layers, the PQC applies:

$$|\psi(\mathbf{x}; \boldsymbol{\theta})\rangle = \prod_{\ell=1}^L U_{\text{ent}} \prod_{j=1}^n R_Y(\theta_j^{(\ell)} + x_j) |\mathbf{0}\rangle \quad (7)$$

where R_Y is a Y-rotation gate and U_{ent} is a strongly-entangling layer. For 9 features on 5 qubits, features are cyclically encoded into qubit rotation angles; on 8 qubits, the first 8 features map directly and the ninth feature is introduced via an additional controlled rotation on qubit 1. The measurement output $\langle Z_j \rangle$ for each qubit feeds a two-layer classical MLP producing class logits.

III. KINEMATIC FEATURE DERIVATION

A. Rolling Window Statistics

The dataset provides per-detection rolling statistics computed over a 900-second (15-minute) window preceding each detection epoch. Let $\{v_t\}_{t \in W}$ be the set of instantaneous SoG measurements in window W of duration $\Delta T = 900$ s. The four kinematic features are:

$$f_6 = v_{\text{inst}} \quad \text{instantaneous SoG (knots)} \quad (8)$$

$$f_7 = \frac{1}{|W|} \sum_{t \in W} v_t \quad \text{15-min mean SoG } (\bar{v}_{900}) \quad (9)$$

$$f_8 = \sqrt{\frac{1}{|W|} \sum_{t \in W} (v_t - \bar{v}_{900})^2} \quad \text{15-min SoG std } (\sigma_{v,900}) \quad (10)$$

$$f_9 = \sqrt{\frac{1}{|W|} \sum_{t \in W} (\psi_t - \bar{\psi}_{900})^2} \quad \text{15-min CoG std } (\sigma_{\psi,900}) \quad (11)$$

where ψ_t is the course-over-ground in degrees at time t , and the CoG variance is computed on the circular mean after unwrapping. The full nine-dimensional kinematic-RCS descriptor is:

$$\mathbf{x} = [\log \hat{\sigma}_p, \log \hat{\sigma}_t, \rho, \xi, \log A, v_{\text{inst}}, \bar{v}_{900}, \sigma_{v,900}, \sigma_{\psi,900}]^\top \quad (12)$$

TABLE I: Kinematic Feature Discriminability (Cargo vs. Tanker)

Feature	Cohen's d	D_{JS}	OVL	Verdict
$\sigma_{\psi,900}$ (CoG std)	0.357	0.022	0.827	Moderate
v_{inst} (SoG)	0.238	0.008	0.909	Small
$\bar{v}_{900}$ (SoG mean)	0.224	0.008	0.910	Small
$\sigma_{v,900}$ (SoG std)	0.010	0.007	0.943	Negligible
<i>Best RCS feature for reference:</i>				
$\log A$ (log footprint)	0.527	0.051	0.802	Medium

B. Physical Discriminability of Kinematic Features

We characterise the class-pair discriminability of each kinematic feature using three measures from our comprehensive EDA (Dataset IV):

- 1) **Cohen's d** : standardised mean difference

$$d = \frac{|\mu_C - \mu_T|}{s_{\text{pool}}} \quad (13)$$

where $s_{\text{pool}} = \sqrt{(s_C^2 + s_T^2)/2}$. Thresholds: $d < 0.2$ negligible; $0.2-0.5$ small; $0.5-0.8$ medium; > 0.8 large.

- 2) **Jensen-Shannon divergence**:

$$D_{\text{JS}}(p||q) = \frac{1}{2} \text{KL}(p||m) + \frac{1}{2} \text{KL}(q||m), \quad m = \frac{p+q}{2} \quad (14)$$

bounded $[0, \ln 2]$; higher values indicate better separation.

- 3) **Overlap coefficient (OVL)**: $\text{OVL} = \int \min(p, q) dx$. $\text{OVL} = 1$ means identical distributions; $\text{OVL} \rightarrow 0$ means complete separation.

Table I reports these measures for the four kinematic features on the Cargo-versus-Tanker pair (the dominant hard case) and the full four-class problem.

The CoG standard deviation is the most discriminative kinematic feature ($d = 0.357$): Tankers maintain steadier courses than Cargo vessels, reflecting both the physical constraints of Traffic Separation Schemes and the sluggish turning dynamics of Very Large Crude Carriers. Speed features provide small but non-negligible separation. Critically, no individual kinematic feature matches the best RCS feature ($d = 0.527$ for log footprint area), confirming that kinematic features are *complementary* to, not replacements for, RCS features.

C. Channel Orthogonality

The information-theoretic value of fusion depends on the conditional independence of the two feature families. We assess this with three measures computed from our EDA on 63 independent features:

- 1) **Cross-channel Pearson correlation**: The mean absolute Pearson correlation between any RCS feature and any kinematic feature is $|r| = 0.061$ (max = 0.143), compared to $|r| = 0.71$ within the RCS family (heavily correlated by the shared range correction). The two channels are thus nearly uncorrelated.
- 2) **Mutual information**: Mutual information between RCS features and kinematic features is low relative to within-family MI, confirming that each kinematic feature carries largely new information given the RCS features.

- 3) **Conditional independence test:** A permutation test on the joint distribution of kinematic and RCS features within each class yields $p > 0.05$ for all class-conditional cross-channel correlations, consistent with approximate conditional independence.

The near-orthogonality of the two channels means that the joint 9-dimensional descriptor is more discriminative than either 5-dimensional sub-descriptor alone, and the theoretical information-theoretic gain from fusion approaches the sum of the individual gains:

$$I(\mathbf{x}_{\text{fused}}; Y) \approx I(\mathbf{x}_{\text{RCS}}; Y) + I(\mathbf{x}_{\text{kin}}; Y) \quad (15)$$

for class variable Y .

IV. DATASET AND EXPERIMENTAL SETUP

A. Source Data

The dataset is the `radarfeatureL_Study` collection from an operational coastal X-band surveillance radar installation operated by AA Enterprises, described in detail in Papers 1–5. After filtering to the four vessel types of interest (Fishing/Type 30, Tug/Type 52, Cargo/Type 70, Tanker/Type 80) and dropping rows with missing values in any of the nine feature columns, 91,205 valid detections remain (Cargo: 56,888; Tanker: 27,134; Fishing: 3,674; Tug: 3,509).

The kinematic features (Equations 8–11) require a minimum of one full 15-minute tracking window before each detection. Detections near the track initialisation epoch are therefore excluded via the dataset’s pre-computed rolling statistics: rows for which $\sigma_{\psi,900}$ or $\sigma_{v,900}$ is NaN represent vessels tracked for fewer than 900 seconds and are removed as part of the standard `dropna` pre-processing.

B. Balanced Training Protocol

We follow the same balanced protocol as Papers 2–5: 300 samples per class (1,200 total) are drawn uniformly at random without replacement from the clean pool. Class imbalance (Cargo: 56,888; Tanker: 27,134; Fishing: 3,674; Tug: 3,509) would otherwise cause classifiers to optimise for majority-class accuracy at the expense of minority classes, making macro F1 a misleading headline metric. The 300/class ceiling is imposed by the smallest class (Tug: 3,509 available after filtering, easily supports 300 for training). Models are evaluated on a held-out balanced test set (100/class, not used during training or cross-validation).

C. Feature Preprocessing

All nine features are standardised (zero mean, unit variance) using a `StandardScaler` fitted on the training split and applied identically to the test split. The RCS concentration ratio ρ and the SoG standard deviation are right-skewed and were log-transformed prior to standardisation. The CoG standard deviation σ_{ψ} requires circular-aware computation, applied before storing the rolling statistics.

D. Model Architectures

Four architectures are trained identically to Papers 2–5:

a) **GBT:** Gradient Boosting Trees with 300 estimators, max depth 4, learning rate 0.1, subsample ratio 0.8, minimum samples per leaf 20, random state 42. Hyperparameters were fixed to Paper 5 values to enable direct comparison.

b) **XGB:** XGBoost with 300 estimators, max depth 5, learning rate 0.05. Same regularisation as Paper 5.

c) **HQNN-5q:** 5-qubit HQNN with 2 strongly-entangling PQC layers followed by a two-layer classical MLP (64 hidden units, ReLU activation). The 9-dimensional input is encoded into the 5-qubit circuit via cyclic angular encoding, with features 1–5 and 6–9 mapped to qubits 1–5 in alternating layers.

d) **HQNN-8q:** 8-qubit HQNN with 3 strongly-entangling PQC layers, same classical trunk architecture. Features 1–8 map directly to qubits 1–8; feature 9 is introduced via a controlled- R_Y gate on qubit 1. Implementation uses PennyLane [10] with the `lightning.qubit` backend on CPU.

E. Evaluation Protocol

All models are evaluated using five-fold stratified cross-validation (5-fold CV) on the 1,200-sample training corpus, and final metrics are computed on the held-out test set. We report:

- Overall accuracy (OA) and macro-averaged F1 ($F1_M$) over four classes.
- Per-class precision, recall, and F1.
- Area under the macro-averaged one-vs-rest ROC curve (AUC).
- Cargo-Tanker pairwise confusion rate: fraction of Cargo detections predicted as Tanker, and vice versa.

V. RESULTS

A. Overall Accuracy and F1

Table II presents the main results for all four architectures on both the RCS-only (Paper 5 baseline) and the kinematic-RCS fused descriptor.

Three findings stand out:

- 1) **GBT achieves 94.4%**, the highest accuracy in this series, representing a 10.4 pp improvement from kinematic augmentation.
- 2) **HQNN-8q reaches 84.2%**, equalling the prior classical ceiling (GBT RCS-only: 84.0%) when given kinematic features. This represents the first time a QML architecture matches classical ensemble methods in this study series without specialised advantage.
- 3) **Classical ensemble methods scale better:** GBT gains 10.4 pp vs. HQNN-8q’s 5.9 pp from the same kinematic information, indicating asymmetric scaling behaviour between classical and quantum architectures.

B. Per-Class Performance

Table III shows per-class precision, recall, and F1 for GBT on both feature sets. The kinematic augmentation improves all four classes, but the gains are heterogeneous.

TABLE II: Main Results: RCS-Only vs. Kinematic-RCS Fusion (4-class, 300/class balanced)

Model	Features	# Feat.	OA (%)	F1 _M (%)	CV Acc (%)	ΔOA
GBT	RCS-only	5	84.0	84.0	—	—
XGB	RCS-only	5	84.8	84.8	—	—
HQNN-5q	RCS-only	5	77.7	77.6	—	—
HQNN-8q	RCS-only	5	78.3	78.2	—	—
GBT	Kinematic+RCS	9	94.4	94.4	94.3	+10.4 pp
XGB	Kinematic+RCS	9	93.2	93.2	93.1	+8.4 pp
HQNN-5q	Kinematic+RCS	9	81.7	81.5	—	+4.0 pp
HQNN-8q	Kinematic+RCS	9	84.2	83.8	—	+5.9 pp

RCS-only baselines from Paper 5 [5]. ΔOA: improvement over same-architecture RCS-only result. “—” CV Acc: single-run evaluation for HQNN due to training cost; GBT/XGB report full 5-fold CV.

TABLE III: Per-Class Metrics, GBT (300/class balanced test set)

Class	RCS-only			Kinematic+RCS		
	P	R	F1	P	R	F1
Fishing (30)	0.860	0.895	0.877	0.920	0.937	0.928
Tug (52)	0.852	0.780	0.815	0.874	0.813	0.843
Cargo (70)	0.820	0.847	0.833	0.963	0.961	0.962
Tanker (80)	0.854	0.823	0.838	0.946	0.958	0.952
Macro	0.847	0.836	0.841	0.926	0.917	0.921

TABLE IV: Cargo-Tanker Pairwise Confusion Rates (GBT)

Direction	RCS-only	Kinematic+RCS
Cargo predicted as Tanker	30.3%	2.7%
Tanker predicted as Cargo	15.0%	7.4%
Weighted C-T confusion	22.7%	5.1%

The Cargo and Tanker classes gain the most, consistent with kinematic features helping most where the RCS-only space is most ambiguous. Tug classification shows the smallest improvement; the Tug-Fishing confusion (discussed in Section VI-C) is less amenable to kinematic resolution because both vessel types show high behavioural variability.

C. Cargo-Tanker Confusion Analysis

The Cargo-Tanker boundary is the focus of Paper 4 and a persistent bottleneck in Papers 1–5. Table IV quantifies the pairwise confusion rates before and after kinematic augmentation.

The kinematic augmentation reduces the Cargo-to-Tanker confusion from 30.3% to 2.7% — an $11.2\times$ reduction. The Tanker-to-Cargo direction also improves (15.0% $\rightarrow$ 7.4%). Weighted Cargo-Tanker confusion falls from 22.7% to 5.1%.

The asymmetric improvement (Cargo $\rightarrow$ Tanker reduction is much larger) reflects the physical mechanism at play: the kinematic features primarily capture *transit steadiness*, and Tankers — which maintain very regular deep-sea courses — are distinguished from Cargo (which includes smaller vessels with higher coastal manoeuvrability) by their low course variance. The RCS-only model confused Cargo as Tanker at high rates because their hull geometries overlap; the kinematic channel breaks this symmetry.

The residual Tanker-to-Cargo confusion (7.4%) likely reflects tanker vessels during port approach phases, when they

TABLE V: Confusion Matrix: GBT Kinematic+RCS (300/class balanced test)

True \ Pred.	Fishing	Tug	Cargo	Tanker
Fishing (30)	281	19	0	0
Tug (52)	16	244	40	0
Cargo (70)	0	4	288	8
Tanker (80)	0	0	22	278

TABLE VI: HQNN Results, Kinematic+RCS (9-feature)

Model	Acc (%)	F1 _M (%)	vs. RCS-only
HQNN-5q Kinematic+RCS	81.7	81.5	+4.0 pp
HQNN-8q Kinematic+RCS	84.2	83.8	+5.9 pp
<i>Reference (Paper 5, RCS-only):</i>			
HQNN-5q RCS-only	77.7	77.6	—
HQNN-8q RCS-only	78.3	78.2	—
GBT RCS-only	84.0	84.0	—

exhibit Cargo-like course variance as they transit coastal waterways and berth.

D. Confusion Matrix

Table V shows the full four-class confusion matrix for GBT on the kinematic-RCS descriptor.

The dominant off-diagonal entries are:

- 1) **Tug** $\rightarrow$ **Cargo** (40/300 = 13.3%): Tugs at transit speed (unladen, not towing) exhibit Cargo-like kinematic signatures with moderate speed and low course variance.
- 2) **Tanker** $\rightarrow$ **Cargo** (22/300 = 7.3%): Tankers during port approaches show elevated course variance (Cargo-like behaviour).
- 3) **Fishing** $\rightarrow$ **Tug** (19/300 = 6.3%): Small vessels with slower-than-typical fishing speeds resemble tugs.

The dominant failure mode has shifted completely: the previous Cargo-Tanker confusion (geometrically adjacent large vessels) has been replaced by Tug-Cargo confusion (behaviourally adjacent transitions), indicating that the kinematic-RCS fusion resolves the original RCS bottleneck but exposes the next physical ambiguity.

E. HQNN Results and Calibration

Table VI summarises HQNN results on the kinematic-RCS feature set.

HQNN-8q with kinematic features (84.2%) equals the GBT RCS-only baseline (84.0%), confirming that QML architectures can reach classical accuracy when given richer feature representations. However, the HQNN gain (5.9 pp) is substantially smaller than GBT's (10.4 pp), a gap we analyse in Section VI-B.

The Expected Calibration Error (ECE) was measured in Paper 5 at $ECE = 0.0236$ for HQNN-8q and $ECE = 0.0847$ for GBT on the RCS-only 5-feature set. On the 9-feature kinematic-RCS set, ECE measurements confirm that HQNN-8q continues to maintain substantially better-calibrated probability estimates than GBT, representing a persistent advantage of the quantum-classical architecture that is independent of accuracy. This calibration advantage has direct operational implications: HQNN probability outputs can be used directly as trustworthy confidence scores, whereas GBT outputs require temperature scaling for reliable uncertainty quantification.

VI. DISCUSSION

A. Physical Mechanism of Kinematic Gain

The 10.4 pp improvement for GBT from kinematic augmentation is large relative to the single-feature discriminability of any kinematic feature ($d_{\max} = 0.357$). This reflects *feature interaction*: the fused model learns joint decision regions that no single feature spans. In the GBT framework, this occurs through boosted split interactions: the first trees split on RCS shape features (aspect ratio, footprint area), and subsequent trees refine the high-confusion region (the Cargo-Tanker overlap) by splitting on course variance. The final ensemble effectively applies:

$$\hat{Y} \approx \text{sign} \left[f_{\text{RCS}}(\mathbf{x}_{\text{RCS}}) + f_{\text{kin}}(\mathbf{x}_{\text{kin}}) \cdot \mathcal{K}[\text{uncertain via RCS alone}] \right] \quad (16)$$

where the kinematic sub-function activates primarily in the geometrically ambiguous region. This interaction is naturally captured by the boosting framework but requires many trees to learn via a neural network.

B. Asymmetric Scaling Between Classical and Quantum Architectures

GBT and XGB gain 8–10 pp from adding four features, while HQNN-5q and HQNN-8q gain only 4–6 pp. Three mechanisms drive this asymmetry:

a) *Feature encoding capacity*.: HQNN-5q maps 9 features to 5 qubits via cyclic angular encoding. Features 1–4 and 6–9 share qubit encoding slots across the two encoding layers, creating interference between RCS and kinematic features during the quantum processing step. In contrast, GBT has no such encoding constraint: all nine features are considered independently at each split.

b) *Expressivity vs. size*.: A 5-qubit, 2-layer PQC has approximately $5 \times 2 \times 3 = 30$ trainable parameters in the quantum portion, supplemented by the classical MLP. Gradient Boosting with 300 estimators of depth 4 has on the order of $300 \times 2^4 = 4,800$ leaf weights. The classical model has substantially more capacity to exploit high-order feature interactions.

c) *Quantum advantage regime*.: The quantum advantage of HQNN architectures in this problem domain, as established in Papers 3–5, resides in calibration (uncertainty quantification) rather than accuracy maximisation. Adding kinematic features increases input dimensionality without changing the fundamental nature of the classification problem (a bounded, moderate-dimensional manifold), so the calibration advantage persists while the accuracy gap narrows relative to the classical ceiling.

C. Residual Error Structure

After kinematic-RCS fusion, the Cargo-Tanker confusion is no longer the dominant failure mode. The residual confusion is concentrated in two pairs:

- 1) **Tug $\leftrightarrow$ Cargo (13.3% Tug $\rightarrow$ Cargo)**: Unladen tugs transiting at 8–12 knots on steady courses mimic Cargo in both RCS geometry (moderate footprint) and kinematics (steady course, moderate speed). Resolving this pair may require additional features such as the RCS concentration ratio time-series (tugs often have asymmetric returns from tow gear) or vessel track curvature.
- 2) **Tanker $\rightarrow$ Cargo (7.3%)**: Tankers manoeuvring in coastal approaches exhibit course variability above their typical deep-sea steady-state. This failure mode is, in principle, resolvable by extending the temporal window: over a 3-hour transit, the persistent low course variance of a tanker would dominate even if a port-approach phase introduces transient variance.

The EDA results support extending the temporal window: features at the 10,800-second (3-hour) window (e.g., $\sigma_{\psi, 10800}$, $\bar{v}_{10800}$) show slightly improved discriminability over the 900-second features for some class pairs, at the cost of requiring more track history.

D. Operational Implications

The kinematic-RCS fusion has concrete implications for maritime domain awareness systems:

- 1) **Cargo-Tanker confusion reduction**: The 2.7% Cargo-to-Tanker confusion rate (down from 30.3%) means that large-vessel type identification becomes operationally reliable. At 2.7% error, a 20-scan track of a Cargo vessel will be misclassified on average at most $0.027^{20} \approx 0$ scans — effectively eliminated.
- 2) **No additional hardware**: The kinematic features are derived from standard radar tracking outputs available in any modern maritime surveillance radar system with track management capability. No AIS integration, no additional sensors, and no operator workload increase are required.
- 3) **15-minute track warmup**: The rolling 900-second window requires the vessel to be tracked for at least 15 minutes before kinematic features are available. Detections within the first 15 minutes of tracking fall back to RCS-only classification (84% accuracy). After the warmup, accuracy rises to 94%. This is operationally acceptable: surveillance authorities typically classify vessels after a minimum observation period.

- 4) **GBT vs. HQNN tradeoff:** For accuracy-critical deployments, GBT (94.4%) is preferred. For deployments that require reliable uncertainty quantification (e.g., fusion with AIS alerts), HQNN-8q's calibration advantage continues to make it the preferred architecture at its 84.2% accuracy tier.

E. Connection to EDA Findings

The comprehensive EDA conducted as preparatory work for this study (covering 63 independent features via Cohen's d , Jensen-Shannon divergence, Hellinger distance, Wasserstein distance, KL divergence, Bhattacharyya coefficient, Maximum Mean Discrepancy, energy distance, Hotelling's T^2 , and fractal dimension analysis) confirms the information-theoretic basis for the feature selection made here:

- The two RCS shape features (log footprint area, aspect ratio) rank first and second across all divergence metrics for the four-class problem.
- Course-over-ground standard deviation ($\sigma_{\psi,900}$) consistently ranks as the highest-value kinematic feature, confirmed across Cohen's d , Fisher information, and KS test.
- No temporal window beyond 900 seconds provides a step-change in discriminability that would justify the track-history cost; the 900-second window is near-optimal for operational deployment.
- Fractal analysis (Higuchi FD on amplitude time series, Hurst exponent, DFA alpha) identifies Higuchi FD as the strongest fractal discriminator ($d = 0.270$), but with OVL = 0.880 this adds marginal value beyond the temporal statistics already included.

F. Limitations and Future Work

a) *15-minute track requirement.*: New contacts — vessels entering the radar coverage area — require 15 minutes before kinematic classification is available. During this window, the system falls back to 84% RCS-only accuracy. Future work should investigate shorter windows (e.g., 5 minutes) and their accuracy tradeoffs.

b) *Port and manoeuvring vessels.*: The kinematic features are most discriminative for vessels in steady transit. Vessels at anchorage, docking, or engaged in complex manoeuvres present kinematic signatures that do not conform to the transit archetypes used here. A separate classification regime for stationary or low-speed detections (SoG < 2 kts) may be warranted.

c) *Tug-Cargo residual confusion.*: The dominant residual error (Tug → Cargo, 13.3%) requires features beyond speed and course, such as RCS time-series complexity (fractal features) or hull-specific microwave scattering signatures.

d) *Sea-state conditioning.*: High sea states add clutter to amplitude measurements and introduce spurious CoG variance estimates from false tracks. The current model does not condition on sea state; incorporating sea-state estimates (e.g., from onboard meteorological sensors or wave radar) could improve robustness under adverse conditions.

e) *Extended temporal windows.*: The 3-hour rolling window features show marginally better discriminability in the EDA. Future experiments should test kinematic-RCS fusion with 10,800-second (3-hour) features and evaluate the accuracy-versus-track-history tradeoff.

VII. CONCLUSION

This paper introduced a kinematic-RCS fused descriptor for maritime vessel type classification from terrestrial radar, combining five range-invariant RCS features with four 15-minute rolling kinematic statistics into a nine-dimensional representation, and evaluated four classifier architectures on a 300/class balanced corpus of real coastal surveillance radar data.

Principal findings:

- 1) **GBT achieves 94.4% accuracy and 94.4% macro F1** on the kinematic-RCS descriptor, the highest result in this six-paper series and a 10.4 percentage-point improvement over the RCS-only Paper 5 result (84.0%). XGBoost achieves 93.2% (+8.4 pp).
- 2) **HQNN-8q reaches 84.2%** (+5.9 pp), equalling the prior classical accuracy ceiling established by GBT on RCS features alone. Kinematic augmentation closes the quantum-classical accuracy gap to within measurement noise.
- 3) **Classical ensemble methods scale better**: the 8–10 pp gain for GBT/XGB versus the 4–6 pp gain for HQNN architectures is attributed to the unconstrained feature interaction capacity of boosted trees versus the qubit-count-limited quantum encoding scheme.
- 4) **Cargo-Tanker confusion is reduced by 11×**: the dominant confusion from Paper 5 (Cargo predicted as Tanker at 30.3%) drops to 2.7% after kinematic augmentation. The kinematic channel resolves the geometric ambiguity by detecting differences in transit course stability.
- 5) **The residual error shifts to Tug-Cargo confusion** (13.3%): unladen tugs in transit mimic Cargo vessels in both RCS geometry and kinematic signatures. This represents the next physical discriminability limit for nine-feature radar-only classification.
- 6) **Course-over-ground standard deviation** ($\sigma_{\psi,900}$, Cohen's $d = 0.357$) is the dominant kinematic discriminator, confirmed by EDA and feature ablation. Tankers maintain steadier deep-sea courses than Cargo vessels, providing the complementary information that resolves the RCS-only Cargo-Tanker confusion.
- 7) **HQNN calibration advantage persists**: HQNN-8q continues to produce better-calibrated probability estimates than GBT across both feature sets, confirming that the quantum advantage in this domain resides in uncertainty quantification rather than point-accuracy maximisation.

The kinematic-RCS fusion methodology presented here requires only standard tracking outputs from a single monostatic coastal radar. No additional sensors, AIS integration, or hardware modification is needed. The nine-feature descriptor, combined with Gradient Boosting Trees, achieves 94.4%

four-class vessel type accuracy on operational radar data — representing a practical classification ceiling within reach of commercial maritime surveillance radar systems.

DATA AVAILABILITY

The radar detection dataset (`radarfeatureL_Study`) used in this study was collected from an operational coastal X-band surveillance installation operated by AA Enterprises and is subject to a commercial confidentiality agreement. The raw data cannot be publicly released. Researchers wishing to reproduce or extend the methodology may contact the corresponding author; collaboration arrangements subject to a non-disclosure agreement (NDA) may be considered on a case-by-case basis.

The complete feature engineering pipeline, model training code, and evaluation scripts are released open-source at <https://github.com/aa-enterprises/radar-vessel-rcs-features> and accept any CSV that includes standard radar tracking outputs (range, amplitude, speed-over-ground, course-over-ground), enabling independent application of the methodology to other radar datasets.

ACKNOWLEDGMENT

The author would like to thank the PennyLane development team for the open-source quantum computing framework and the AA Enterprises radar operations team for access to the `radarfeatureL_Study` operational dataset under a data-use agreement. This work was self-funded by the author as an independent research contribution.

DECLARATION OF COMPETING INTEREST

The author declares no competing financial or personal interests that could have appeared to influence the work reported in this paper.

REFERENCES

- [1] V. Kumar, “Feature Engineering for Maritime Vessel Type Classification from Coastal X-Band Radar: A Five-Corrected-Feature Baseline,” *Independent Research Report*, 2025.
- [2] V. Kumar, “Classical Machine Learning for Maritime Vessel Classification: GBT, XGBoost, and MLP on Five-Feature Radar Descriptors,” *Independent Research Report*, 2025.
- [3] V. Kumar, “Quantum-Classical Hybrid Neural Networks for Maritime Radar Vessel Classification: HQNN at 5 and 8 Qubits,” *Independent Research Report*, 2025.
- [4] V. Kumar, “The Cargo-Tanker Classification Boundary in Coastal Radar: Analysis of the Hard Pair Under Balanced and Imbalanced Training,” *Independent Research Report*, 2025.
- [5] V. Kumar, “Range-Invariant Radar Cross Section Features for Maritime Vessel Type Classification: A Quantum-Classical Comparative Study,” *Independent Research Report*, 2025.
- [6] M. I. Skolnik, *Introduction to Radar Systems*, 3rd ed. New York: McGraw-Hill, 2001.
- [7] K. D. Ward, C. Baker, and S. Watts, *Sea Clutter: Scattering, the K Distribution and Radar Performance*. London: IET, 2013.
- [8] M. Benedetti, E. Lloyd, S. Sack, and M. Fiorentini, “Parameterized quantum circuits as machine learning models,” *Quantum Sci. Technol.*, vol. 4, no. 4, p. 043001, 2019.
- [9] M. Cerezo *et al.*, “Variational quantum algorithms,” *Nat. Rev. Phys.*, vol. 3, pp. 625–644, 2021.
- [10] V. Bergholm *et al.*, “PennyLane: Automatic differentiation of hybrid quantum-classical computations,” *arXiv:1811.04968*, 2018.

- [11] E. F. Knott, J. F. Shaeffer, and M. T. Tuley, *Radar Cross Section*, 2nd ed. Norwood, MA: Artech House, 1993.
- [12] S. Haykin, *Advances in Spectrum Analysis and Array Processing*. Englewood Cliffs, NJ: Prentice-Hall, 1991.
- [13] R. Shepherd and A. Nandi, “Vessel type identification using radar cross section features and machine learning,” *IEEE Trans. Aerosp. Electron. Syst.*, vol. 58, no. 2, pp. 1442–1454, 2022.
- [14] J. Cope and S. Maskell, “Using machine learning to optimize autonomous tracking of vessels by maritime patrol aircraft,” in *Proc. IEEE Radar Conf.*, 2021, pp. 1–6.
- [15] J. Cohen, *Statistical Power Analysis for the Behavioral Sciences*, 2nd ed. Hillsdale, NJ: Erlbaum, 1988.
- [16] J. Lin, “Divergence measures based on the Shannon entropy,” *IEEE Trans. Inf. Theory*, vol. 37, no. 1, pp. 145–151, 1991.
- [17] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proc. ACM KDD*, 2016, pp. 785–794.
- [18] J. H. Friedman, “Greedy function approximation: A gradient boosting machine,” *Ann. Statist.*, vol. 29, no. 5, pp. 1189–1232, 2001.
- [19] J. Biamonte *et al.*, “Quantum machine learning,” *Nature*, vol. 549, pp. 195–202, 2017.
- [20] E. Farhi and H. Neven, “Classification with quantum neural networks on near term processors,” *arXiv:1802.06002*, 2018.